

Assignment 1: An Automated Warehouse Sorting Robot

Task 1: PEAS Specification

Performance Measure: To measure success, the following four metrics are used:

1. Total number of items correctly sorted and delivered to the right station per hour.
2. Minimization of the package damage rate (number of dropped or crushed items).
3. Energy efficiency, measured by battery power used per working shift.
4. Number of safety incidents, such as collisions with objects, humans, or other robots.

Environment: The agent acts on the physical warehouse floor. Key external factors include shelving units, charging stations, changing lighting conditions, floor friction, and dynamic obstacles like human workers and other moving robots.

Actuators: The robot changes its world using mechanized driving wheels for moving, pneumatic lifting arms for picking up bins, LED status lights to signal humans, and a built-in speaker for warning beeps.

Sensors: The robot perceives its state using high-resolution RGB and Infrared sensors for vision, LiDAR for mapping the space, RFID readers to identify package details, and tactile pressure sensors on its lifting arms to feel its grip on items.

Task 2: Environment Classification

- **Fully Observable vs. Partially Observable:** Partially Observable, because the robot's sensors cannot see the entire warehouse layout around corners or see the specific contents hidden inside sealed boxes.
- **Single-agent vs. Multi-agent:** Multi-agent, because the robot must interact with and avoid other autonomous sorting robots and human workers sharing the exact same workspace.
- **Deterministic vs. Stochastic:** Stochastic, because unpredictable events happen, such as packages slipping, humans walking into its path, or sensors failing.
- **Episodic vs. Sequential:** Sequential, because a current action (like taking a specific aisle) directly affects the robot's future choices and its overall delivery time.
- **Static vs. Dynamic:** Dynamic, because the warehouse environment is constantly changing while the robot decides what to do, with other machines and people constantly moving around it.
- **Discrete vs. Continuous:** Continuous, because the robot's physical movement, speed, and steering angles happen smoothly over continuous time and physical space.

Task 3: Critical Analysis

1. Greatest Technical Challenge: The greatest technical challenge for an AI engineer is dealing with the Dynamic and Multi-agent nature of the warehouse. Why? The robot must constantly recalculate its paths in real-time to avoid crashes. Predicting the sudden, unpredictable

movements of human workers and syncing movements with dozens of other robots requires highly complex, fast-acting algorithms to ensure everyone stays safe while keeping the work fast.

2. Utility Function and Trade-offs: To handle the trade-off between Speed and Safety, the agent could use a Utility Function like: $Utility = (Positive\ Weight * Delivery\ Speed) - (Negative\ Weight * Collision\ Risk)$. If the weight of this safety utility was doubled, the agent's behavior would change by becoming much more cautious. The robot would drive at slower speeds, take wider and safer paths around human workers, and pause longer at busy intersections. This would reduce the overall speed of package sorting but greatly increase warehouse safety.

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ROLL NO: 2K23/CSE/2

COURSE: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

TOPIC: PEAS FRAMEWORK AND TASK ENVIRONMENT ANALYSIS

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